

Likelihood Ratio Tests

- The last section presented an inference for estimation and confidence intervals based on likelihood theory. This section will present corresponding inference for testing hypotheses.
- Assume $X_1, \dots, X_n$ be iid from pdf $f(x; \theta)$ for $\theta \in \Omega$. Consider the two-sided hypotheses:

$$H_0 : \theta = \theta_0, \quad \text{vs.} \quad H_1 : \theta \neq \theta_0,$$

where θ_0 is specified value.

- Likelihood and log likelihood:

$$L(\theta) = \prod_{i=1}^n f(X_i; \theta)$$

$$l(\theta) = \sum_{i=1}^n \log f(X_i; \theta)$$

- $\hat{\theta}$ is mle of θ , which maximizes $l(\theta)$,

$$\hat{\theta} = \arg \max_{\theta \in \Omega} l(\theta).$$

- Consider the ratio (test statistics)

$$\Lambda = \frac{L(\theta_0)}{L(\hat{\theta})}.$$

- Note that $\Lambda \leq 1$ since $\hat{\theta}$ is the maximum of $L(\theta)$. If H_0 is true, Λ should be close 1; while if H_1 is true, Λ should be smaller.
- For a specified significance level α , this leads to the intuitive decision rule

Reject H_0 in favor of H_1 if $\Lambda \leq c$,

where $\alpha = \mathbb{P}_{\theta_0}(\Lambda \leq c)$.

- Likelihood ratio test for the exponential distribution: Suppose $X_1, \dots, X_n$ are iid with pdf $f(x; \theta) = \theta^{-1} \exp\{-x/\theta\}$, for $x, \theta > 0$. The likelihood function is

$$L(\theta) = \theta^{-n} \exp\{-(n/\theta)\bar{X}\}.$$

The mle is $\bar{X}$.

- Likelihood ratio test statistic:

$$\Lambda = e^n \left(\frac{\bar{X}}{\theta_0}\right)^n \exp\{-n\bar{X}/\theta_0\} = e^n g\left(\frac{\bar{X}}{\theta_0}\right),$$

where $g(t) = t^n e^{-nt}$.

Decision rule is to reject H_0 if $\Lambda \leq c$,

$$\Lambda \leq c, \text{ if and only if } \frac{\bar{X}}{\theta_0} \leq c_1 \text{ or } \frac{\bar{X}}{\theta_0} \geq c_2.$$

Note that under the null hypothesis H_0 , the statistics $(2/\theta_0) \sum_{i=1}^n X_i$ has χ^2 distribution with degrees of freedom $2n$. So the decision rule in a level α test is

Reject H_0 if $(2/\theta_0) \sum_{i=1}^n X_i \leq \chi_{1-\alpha/2}^2(2n)$ or $(2/\theta_0) \sum_{i=1}^n X_i \geq \chi_{\alpha/2}^2(2n)$

- Likelihood ratio test for the mean of a normal pdf: Say $X_1, \dots, X_n$ is a random sample from a $N(\theta, \sigma^2)$ distribution and σ^2 is known. Consider the test,

$$H_0 : \theta = \theta_0, \quad \text{vs.} \quad H_1 : \theta \neq \theta_0.$$

The likelihood function is

$$L(\theta) = \left(\frac{1}{2\pi\sigma^2}\right)^{n/2} \exp\{-(2\sigma^2)^{-1} \sum_{i=1}^n (x_i - \bar{x})^2\} \exp\{-(2\sigma^2)^{-1} n(\bar{x} - \theta)^2\}.$$

The mle is $\hat{\theta} = \bar{X}$.

$$\Lambda = \frac{L(\theta_0)}{L(\hat{\theta})} = \exp\{-(2\sigma^2)^{-1}n(\bar{X} - \theta_0)^2\}.$$

Then $\Lambda \leq c$ iff $-2 \log \Lambda \geq -2 \log c$. However,

$$-2 \log \Lambda = \left(\frac{\bar{X} - \theta_0}{\sigma/\sqrt{n}} \right)^2$$

has $\chi^2(1)$ distribution under H_0 . Thus the decision rule is to reject H_0 and accept H_1 when

$$-2 \log \Lambda = \left(\frac{\bar{X} - \theta_0}{\sigma/\sqrt{n}} \right)^2 \geq \chi^2_{\alpha}(1).$$

Asymptotic Distribution of the Test Statistic Under H_0

- In the last two examples we are able to get the test in closed form. Often this is impossible.
- Theorem 6.3.1. Assume the same regularity conditions for Theorem 6.2.2. Under the null hypothesis, $H_0 : \theta = \theta_0$,

$$-2 \log \Lambda \xrightarrow{D} \chi^2(1).$$

- Define the test statistic $\chi_L^2 = -2 \log \Lambda$. The decision rule

Reject H_0 in favor of H_1 if $\chi_L^2 \geq \chi_\alpha^2(1)$

This test has the asymptotic level α .